

Control Systems in a Nutshell

A PowerPoint2Beamer demo

Abolfazl Mohebbi

September 19, 2026

Outline

1 Introduction

2 Analysis

3 Design

Feedback control

- A controller compares the **output** with the *reference*
 - Proportional, integral and derivative actions
 - Tuned from a model or from experiments
- Too much gain makes the loop **unstable!**
- A well-tuned loop is **fast and stable**

Transfer functions

The Laplace transform turns differential equations into algebra:

$$F(s) = \int_0^{\infty} f(t)e^{-st}dt$$

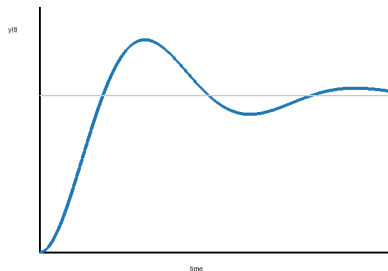
First-order and second-order systems:

$$G(s) = \frac{K}{\tau s + 1}$$

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Step response

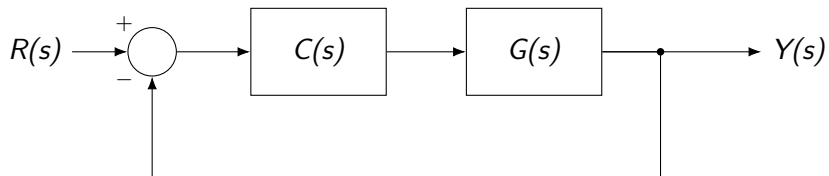
- The output after a sudden change of the input
 - Rise time
 - Overshoot
 - Settling time
 - Steady-state error



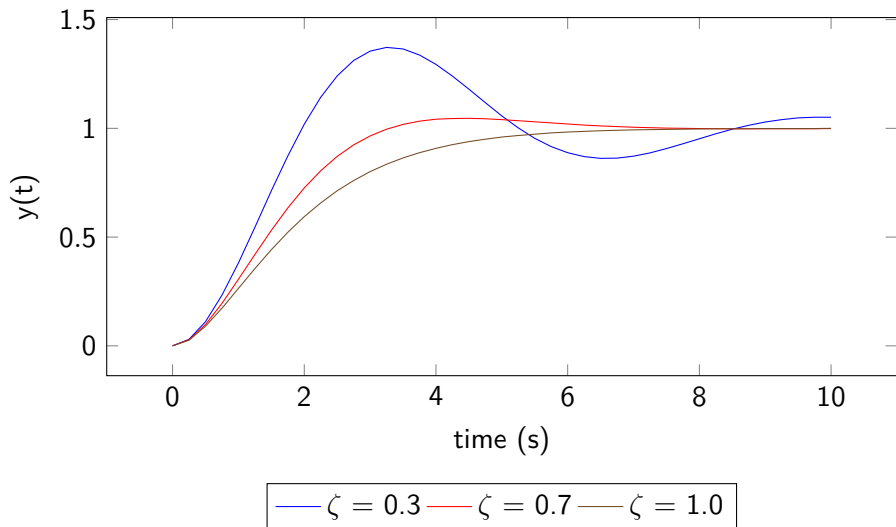
Ziegler–Nichols tuning

Controller	K_p	T_i	T_d
P	$0.5 K_u$	—	—
PI	$0.45 K_u$	$T_u / 1.2$	—
PID	$0.6 K_u$	$T_u / 2$	$T_u / 8$

Closed-loop block diagram



Response to a step input



Designing a controller

- 1. Model the system

Designing a controller

- 1. Model the system
- 2. Choose the specifications

Designing a controller

- 1. Model the system
- 2. Choose the specifications
- 3. Tune the controller

Designing a controller

- 1. Model the system
- 2. Choose the specifications
- 3. Tune the controller
- 4. Simulate and verify